

THE UNIVERSITY OF LAHORE

Department of Computer Science & IT

Faculty of Information Technology

Final Year Project Proposal

DATE	Day	Month	Year
	2	2	05
	0	5	20
	2	0	25

PROJECT TITLE:	Dark sea
KEY WORDS:	Horror game, Audio-Based Navigation, AI
DOMAIN OF THE PROJECT:	Game Development, AI, 3D Sound design
SUPERVISOR'S NAME:	Sir Shahid Yousaf

STUDENT INFORMATION				
Write down the detail of all group members in BLOCK LETTERS ONLY.				
Sr	Student ID	Name	Email	Mobile
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PROBLEM STATEMENT
Most horror games use visuals like dark graphics and jump scares to create fear. Because of this, blind or visually impaired players cannot enjoy them. There are very few horror games that focus on sound-based gameplay. This creates a gap in the gaming world for accessible horror experiences. Our project, “Dark Sea,” solves this problem by using only sound and vibration to guide the player. It gives both blind and sighted users an equal chance to enjoy a scary and immersive game.

EXECUTIVE SUMMARY
“Dark Sea” is a survival horror game set in a deep-sea environment. Players use sound to explore, survive, and avoid dangerous creatures. The game includes AI enemies that copy real voices to confuse players. There are no visuals required, so blind and sighted users can enjoy the same level of fear and excitement. The game will be developed using Unreal Engine 5 and an audio system like FMOD or Wwise.

INTRODUCTION

This game is about surviving in deep underwater area using sounds. “Dark Sea” is a psychological horror survival game that takes place in a deep-sea environment. Unlike most games, it is made for both sighted and visually impaired players. Players use sound instead of sight to move around, survive, and avoid danger. The game includes sonar-based navigation, where players can "ping" sound waves to detect their surroundings.

As players go deeper into the sea, the environment becomes confusing and scary. The enemies in the game, called **Mimics**, are powered by AI and can copy human voices even voices that sound like your friends to trick you. The game creates fear through sound, not visuals, making it inclusive and unique.

COMPETITORS/COMPETITIVE ANALYSIS

The competitors of our projects are

1.Perception

A horror game that uses echolocation, but is mostly visual

2.BlindSide

A 3D audio-only adventure game

3. The Vale: Shadow of the Crown

An audio-only RPG for blind players

Competitive Advantage of Dark Sea:

- It gives equal gameplay to both blind and sighted users
- Uses realistic 3D audio and voice-mimicking AI
- Has a survival horror theme, which is rare in accessible games
- Offers a deeper and more emotional experience with psychological fear

OBJECTIVES

The objectives of our projects are:

- Make a horror game that works with sound only
- Let blind and sighted players enjoy the game equally
- To use AI that can copy human voices and create fear through sound
- To include accessibility features like 3D audio and vibration feedback.
- Create a deep-sea environment with realistic sound and survival mechanics

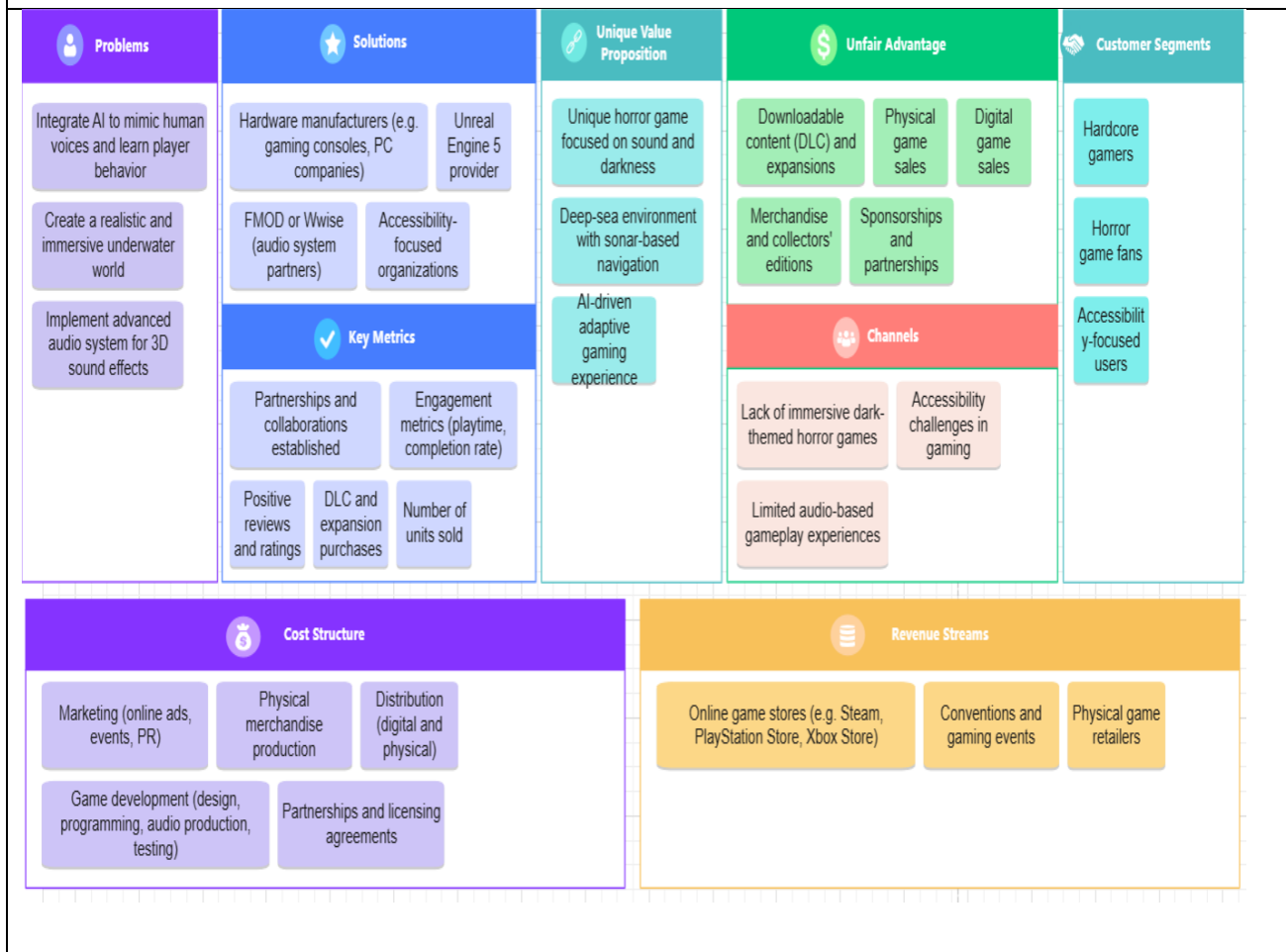
MOTIVATION

Blind gamers usually don't have access to horror games. We want to change that. We also want to challenge ourselves by working on AI, audio, and accessibility all in one unique game. We want to include visually impaired people in horror gaming. Most games are not designed for them, especially in the horror genre. This project is a chance to build something new and inclusive, and also test our technical skills with AI and sound. We are also motivated to create a game that brings fear through sound alone, which is a creative and exciting challenge. It gives us an opportunity to explore accessibility, voice technology, and immersive audio design all growing fields in tech and gaming.

FEATURES OF PROJECT

- **Sonar Navigation:**
Players use sound waves (sonar) to explore the deep-sea environment.
- **Voice-Mimicking AI:**
Enemies can copy human voices to trick and confuse players.
- **3D Spatial Audio:**
Sounds come from different directions to help players understand the environment.
- **Oxygen Survival System:**
Players must manage their oxygen level to stay alive underwater.
- **Psychological Horror Effects:**
The game becomes more confusing and scary as players go deeper.
- **Vibration Feedback:**
Vibrations are used to alert players about danger without needing visuals.
- **Accessible Design:**
The game is fully playable by both sighted and visually impaired users.
- **Adaptive Enemy Behavior:**
Enemies learn from how the player moves and respond more intelligently.
- **Minimal Visuals:**
The game focuses on audio, so visuals are simple or optional.

ARCHITECTURAL DESIGN



IMPLEMENTATION TOOLS AND TECHNIQUES

Tools:

- Unreal Engine 5** – For game development and environment design
- FMOD or Wwise** – For realistic 3D audio and sound effects
- Audacity** – To record and edit voice and sound clips
- GitHub** – For version control and team collaboration
- Trello** – For managing project tasks and timelines

Techniques:

- Sound-Based Navigation** – Using sonar pings and 3D audio for movement
- Voice Mimicry using AI** – AI learns and copies voices to scare players
- Environmental Audio Mapping** – Designing soundscapes to replace visuals
- Accessibility Integration** – Adding vibration and spatial sound for blind users
- Survival Mechanics** – Oxygen management and time-limited exploration

PROJECT PLAN

May25-Jun	Jun-Aug	Aug-Oct	Oct-Dec	Dec-Jan	Jan-Mar	Mar-Apr	Apr-may26
Planning & Research							
	Environment Setup						
		Sonar Navigation System					
			AI Development (Mimics)				
				Survival Mechanics			
					Accessibility Features		
						Testing and Feedback	
							Finalization and Deployment

Supervisor's Signature:

FYP Proposal Evaluation

(To be filled by students)

PROJECT TITLE:	E-Voting System Using Blockchain
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STUDENT INFORMATION				
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Sr.	Student ID	Name	Email	Mobile
1.				
2.				
3.				

FOR EVALUATOR'S ONLY			
Remarks:	☐ Accepted	☐ Accepted with Minor changes	☐ Rejected

[illegible]

Reviewed By:

Name: _____

Signature: _____

Day Month Year

DATE			-			-				
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